Project Documentation

Vehicle Parking App

1. Author

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2. Description

This project is a Vehicle Parking Application developed using Flask for the backend, SQLite for the database, and HTML/CSS/Bootstrap with Jinja2 templating for the frontend.

It supports two roles: **Admin** and **User**. Admins can manage parking lots and spots, while users can book, release, and view their reservations. Each user type has a customized dashboard with relevant options.

3. Technologies Used

Backend

- Flask – Web framework
- Flask-SQLAlchemy – ORM
- SQLite – Lightweight database
- Jinja2 – Template rendering

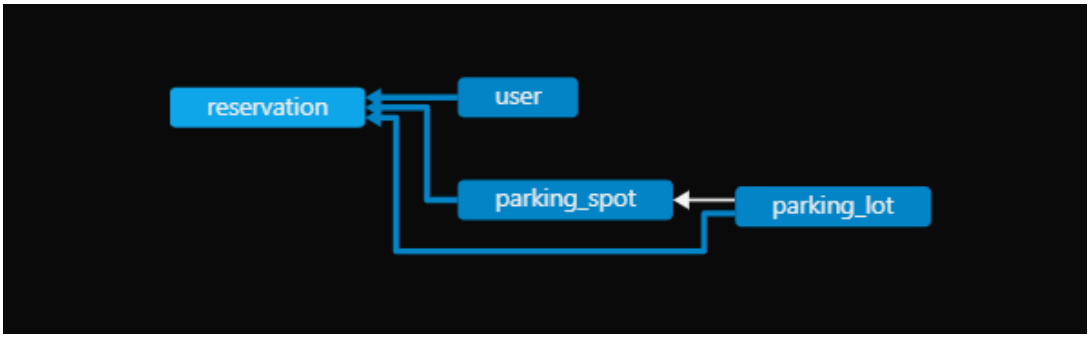
Frontend

- HTML – Page structure
- CSS and Bootstrap – Styling and layout
- JavaScript – Minor interactivity (e.g., form confirmations)

4. Database Schema Design

- **Admin Table** – Stores admin details
- **User Table** – Stores user details
- **ParkingLot Table** – Contains parking lot info
- **ParkingSpot Table** – Contains individual spot status and association with lot
- **Booking Table** – Tracks which user booked which spot and when

5. Database Schema Diagram



6. Architecture and Structure

The application follows a modular structure:

`app.py`	– Main Flask application entry point
`/templates/`	– HTML pages with Jinja2 templating
`/static/`	– CSS and image assets
`/controllers/auth.py`	– Handles login, logout, registration
`/controllers/admin_routes.py`	– Admin dashboard and parking management
`/controllers/user_routes.py`	– User dashboard and booking features
`/models/models.py`	– SQLAlchemy models for all tables
`README.md`	– Setup instructions and usage

7. Implemented Features

- Predefined Admin login and dynamic User registration/login
- Role-based dashboard redirection after login
- Admin:
  - Add/Edit/Delete parking lots and spots
  - View bookings and user activity
- User:
  - View available spots
  - Book or release a parking spot
  - View their booking history
- Profile edit pages
- Responsive UI using Bootstrap
- Dark mode using javascript

8. Video

Project Demo: https://drive.google.com/file/d/1CJSmTAcMvXFaNVxnmLwYlEnCcXLVSTNa/view?usp=sharing